Ishaan Shankar

Undergraduate Student — Avionics — Robotics — Space Technology

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Profile

Third-year undergraduate student at the Indian Institute of Space Science and Technology specializing in Avionics, with a strong interest in autonomous and space systems. Passionate about solving space-related challenges by integrating robotic dynamics, embedded systems, and control theory. Hands-on experience in satellite communication systems, quadcopter avionics, computer vision, and digital communication protocols.

Education

B.Tech in Avionics Engineering Expected Graduation: 2028
Indian Institute of Space Science and Technology (IIST), India
CGPA: **8.72** — Last Semester GPA: **8.83**

Academic Achievements

- Qualified JEE Advanced with AIR 6290.
- Secured admission into IIST (under Department of Space, ISRO).
- Served as a Class Committee Member.

Projects

Winner of ORION Hackathon 2K26 10 Oct 2026 - 11 Oct 2026

- Part of Team Varuna, which secured first place in ORION UAV Hackathon 2k26 conducted by P.E.G.A.S.U.S. UAV Club, IIST.
- Ship-to-Shore & Island Supply Drone: Design a drone to supply a 5kg payload to a shore/island, 15km away from a vessel, faring 12m/s winds, and landing/dropping the payload onto a moving platform of speed 2m/s.
- Worked on creating an SITL VTOL drone in Gazebo while also creating a custom Gazebo world that would simulate the winds.

GPS-denied Navigation for a quadcopter Ongoing

- Constructing a pipeline network to run visual SLAM on RPi5
- Currently running simulations, incorporating Gazebo and RTAB-Map along with optimizations.

Road Anomaly Detection — Bharat AI-SOC Student Challenge (2026) Jan 2026 – Mar 2026

- Developed a computer vision pipeline using Raspberry Pi for pothole and obstacle detection.
- Implemented deep learning models for real-time classification and inference.

UART Protocol Implementation — Course Project Aug 2025 – Dec 2025

- Designed and verified UART transmitter and receiver modules.
- Implemented the design using Verilog HDL and tested using simulation tools.

Student CubeSat Communication Systems May 2025 – Oct 2025

- Worked on the design and analysis of satellite communication subsystems.
- Contributed to link budget analysis, signal integrity considerations, and subsystem integration.

Internship

Summer Internship project - DRDL, Hyderabad May 2026 – July 2026

- Studied the design of a three-loop autopilot for a tactical missile.
- Designed a notch filter for the three-loop autopilot to attenuate the missile's vibration frequencies.
- Navigated through missile flight dynamics and its complications.

Technical Skills

Simulation / Design Tools: MATLAB, Simulink, LTspice, KiCad, AutoCAD
Programming: Python, C++, Verilog (HDL)
Domains: Analog Circuit Design, VLSI Design, Embedded Systems, Control Systems
Computer Vision: OpenCV, Image Processing
Hardware Platforms: Arduino, Raspberry Pi

Relevant Coursework

Signals and Systems, Electromagnetic Theory, Analog Circuits, Digital Circuits, Control Systems, Embedded Systems, Semiconductor Devices, Sensor Electronics

Leadership & Extracurricular Activities

- Club Head — **The Astoun** (Dance Club of IIST).
- Member of Publicity Team for Technical and Cultural Festivals at IIST.
- Participant in Arduino Hackathon organized during Technical Fest at IIST.

Interests

Autonomous Systems, Space Robotics, Satellite Systems, Embedded AI, Control Algorithms, Space Exploration